

Medicare Cardiovascular Spending Variation by Anxiety Status

Davey Johnson

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Abstract

We find that Medicare beneficiaries with anxiety face consistently higher healthcare spending across major cardiovascular conditions, including hypertension, coronary artery disease (CAD), and atrial fibrillation (AFib). In unadjusted comparisons, patients with anxiety spend approximately \$2,600–\$3,000 more annually than those without anxiety, equivalent to 44.6%–63.1% higher spending. After adjusting for demographic and clinical characteristics, spending remains 7.8%–12.4% higher among patients with anxiety, indicating that the relationship is not fully explained by underlying health differences. Two-part models show that this gap is driven primarily by outpatient use rather than inpatient care. State-level spending gaps are positive across all states analyzed. These findings suggest that anxiety represents an independent and consistent driver of healthcare use among cardiovascular patients and may be an important target for improving efficiency in Medicare.

Introduction

Cardiovascular diseases such as hypertension, coronary artery disease (CAD), and atrial fibrillation (AFib) are among the most common and costly conditions in the Medicare population. These conditions require ongoing management, including taking medications as prescribed, routine monitoring, and frequent interactions with the healthcare system. Because of this, even relatively small differences in healthcare use patterns can result in significant differences in total spending.

Mental health conditions, particularly anxiety, which is characterized by persistent worry, fear, or nervousness, are increasingly recognized as influencing how patients interact with the healthcare system. Patients with anxiety may be more likely to seek care for symptoms, request additional testing, or require more frequent follow-up visits. Anxiety may also affect how

patients perceive symptoms, follow treatment plans, and react to uncertainty about their health. These behavioral pathways can translate into higher levels of healthcare use, particularly in outpatient settings.

However, it is difficult to determine whether higher spending among patients with anxiety reflects an independent effect of anxiety or simply differences in underlying health status. Patients with anxiety may have more comorbid conditions, greater disease severity, or other characteristics that drive higher healthcare use. If this is the case, then observed spending differences may not actually reflect a direct effect of anxiety itself.

Distinguishing between these explanations is important for both policy and care delivery. If spending differences are driven primarily by patient characteristics, interventions should focus on improving clinical management of high-risk patients. If anxiety independently contributes to higher healthcare use, then addressing mental health may represent an opportunity to improve both efficiency and patient outcomes.

In this analysis, we examine the relationship between anxiety and Medicare spending across three major cardiovascular conditions. We present unadjusted spending differences, estimate adjusted effects controlling for patient characteristics, examine what is causing these differences, and assess whether these patterns are consistent across the United States.

Methods

Data and Study Population

We analyzed CMS DE-SynPUF Medicare claims data (Sample 1) for beneficiaries with hypertension, CAD, or AFib. These conditions were selected because they are common and account for a meaningful share of Medicare spending. Patients were then categorized based on the presence or absence of anxiety using diagnosis codes recorded in the claims data.

We examined total annual spending as the primary outcome, along with inpatient and outpatient spending components. Total spending includes all reimbursed services during the observation period and provides a broad measure of healthcare use.

Analytical Approach

We conducted four analyses to examine different dimensions of the relationship between anxiety and spending.

First, we estimated condition-specific Gamma generalized linear models (GLMs) with a log link to model total spending. These models controlled for age group, sex, race, and comorbidity count. This approach allows us to estimate adjusted differences in spending between patients with and without anxiety while accounting for observable patient characteristics.

Second, we implemented two-part models to better understand use patterns. The first part used logistic regression to estimate the probability of any spending, capturing whether patients engage with care. The second part used another Gamma GLM to estimate spending among patients with positive expenditures, capturing how much was spent once patients had received care.

Third, we tested for heterogeneity in the anxiety effect across cardiovascular conditions and age groups using interaction terms. This analysis helped determine whether the relationship between anxiety and spending varied across patient subgroups.

Finally, we conducted a geographic analysis by calculating state-level spending gaps between patients with and without anxiety and examining their distribution across states.

Results

Unadjusted Spending Differences

In Figure 1, we show the unadjusted mean annual spending by anxiety status across cardiovascular health conditions. Patients with anxiety have consistently higher spending across all conditions, with differences ranging from approximately \$2,600 to \$3,000 per year, corresponding to 44.6% to 63.1% higher spending. In hypertension we find the largest spending gap, with slightly smaller but still significant differences for CAD and AFib. These raw differences represent large relative increases in spending and provide an initial indication that anxiety is associated with higher healthcare use. However, because these estimates do not account for differences in patient characteristics, they may reflect underlying variation in health status rather than a direct effect of anxiety itself.

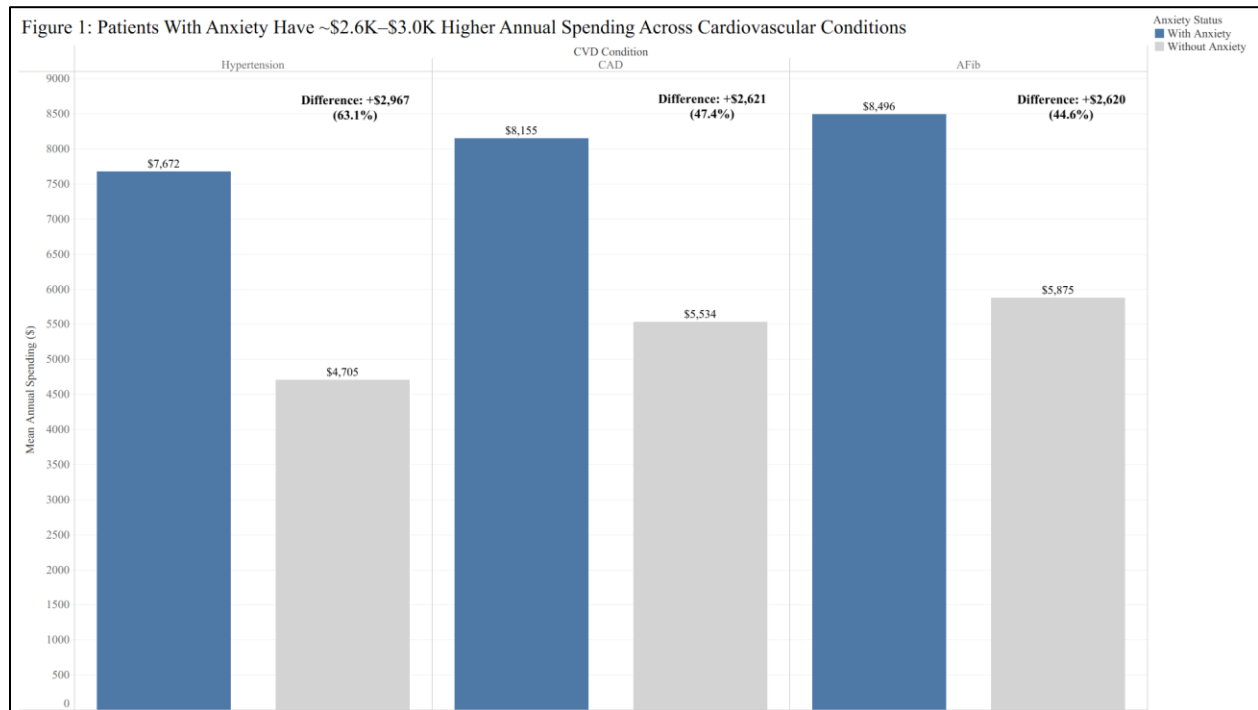


Figure 1: Unadjusted mean annual Medicare spending by anxiety status and cardiovascular condition. Patients with anxiety have substantially higher annual spending across all cardiovascular conditions. Differences range from approximately \$2,600 to \$3,000 per beneficiary. These estimates are unadjusted and reflect raw differences in spending without controlling for demographic or clinical characteristics.

Adjusted Spending Differences

In Figure 2, we present the adjusted mean annual spending estimates controlling for age group, sex, race, and comorbidity count. Patients with anxiety continue to have higher spending across all three cardiovascular conditions after adjustment. The adjusted dollar gaps range from approximately \$544 to \$725 per beneficiary, corresponding to 7.8% to 12.4% higher spending. The largest adjusted effect is observed for hypertension, followed by CAD and AFib. Although these differences are smaller than in the unadjusted comparisons, the presence of both positive dollar gaps and positive percentage effects indicates that anxiety remains independently associated with increased healthcare spending. The smaller size of effect after adjustment suggests that some of the raw differences are explained by patient characteristics, but a significant portion still remains.

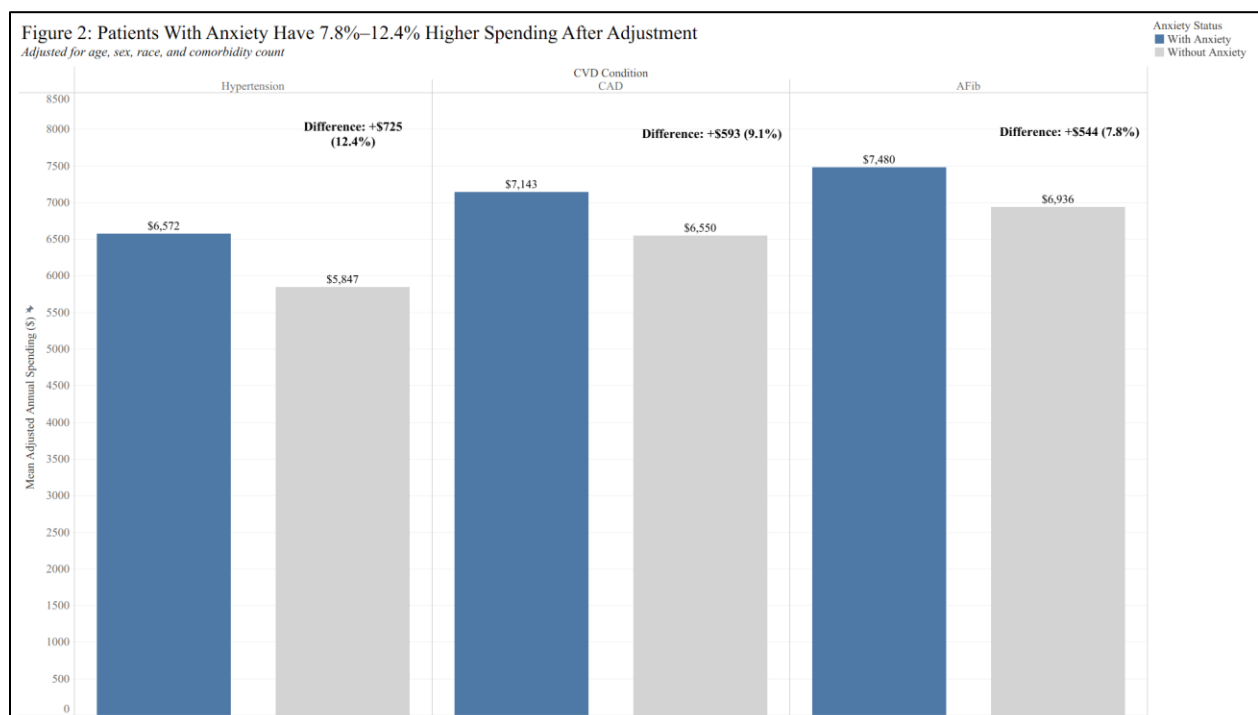


Figure 2: Adjusted mean annual Medicare spending by anxiety status and cardiovascular condition. After controlling for age group, sex, race, and comorbidity count using Gamma generalized linear models, patients with anxiety continue to have higher spending across all cardiovascular conditions. Adjusted differences range from approximately \$544 to \$725 per beneficiary (7.8%–12.4% higher spending).

Although the size of the differences is smaller than in unadjusted comparisons, the persistence of statistically significant effects indicates that anxiety independently contributes to higher spending. Specifically, after adjustment, the cost ratio for anxiety was 1.124 (95% CI: 1.094–1.155) for hypertension, 1.090 (95% CI: 1.061–1.120) for CAD, and 1.078 (95% CI: 1.048–1.110) for AFib, all with p-values < 0.001. The reduction in magnitude suggests that part of the observed gap is explained by patient characteristics, but a meaningful portion remains after adjustment.

The differences in effect size across conditions may reflect distinctions in how anxiety influences disease management. For example, hypertension care often depends heavily on outpatient monitoring and patient engagement, which may make the effects of anxiety worse.

Healthcare Use Patterns

Figure 3 shows the adjusted spending differences by care setting and cardiovascular condition, expressed in dollar terms as the difference between patients with anxiety and those without anxiety. Across hypertension, CAD, and AFib, outpatient spending is consistently about \$168 to \$169 higher among patients with anxiety, while inpatient differences are small and slightly negative. This indicates that the overall anxiety spending gap is concentrated in outpatient care rather than inpatient use. In other words, patients with anxiety do not appear to drive meaningfully higher inpatient spending, but they do show consistently higher outpatient spending. This pattern suggests that anxiety affects how patients interact with the healthcare system through higher frequency or more intensive outpatient care, such as follow-up visits,

monitoring, or diagnostic testing, rather than through greater use of inpatient care.

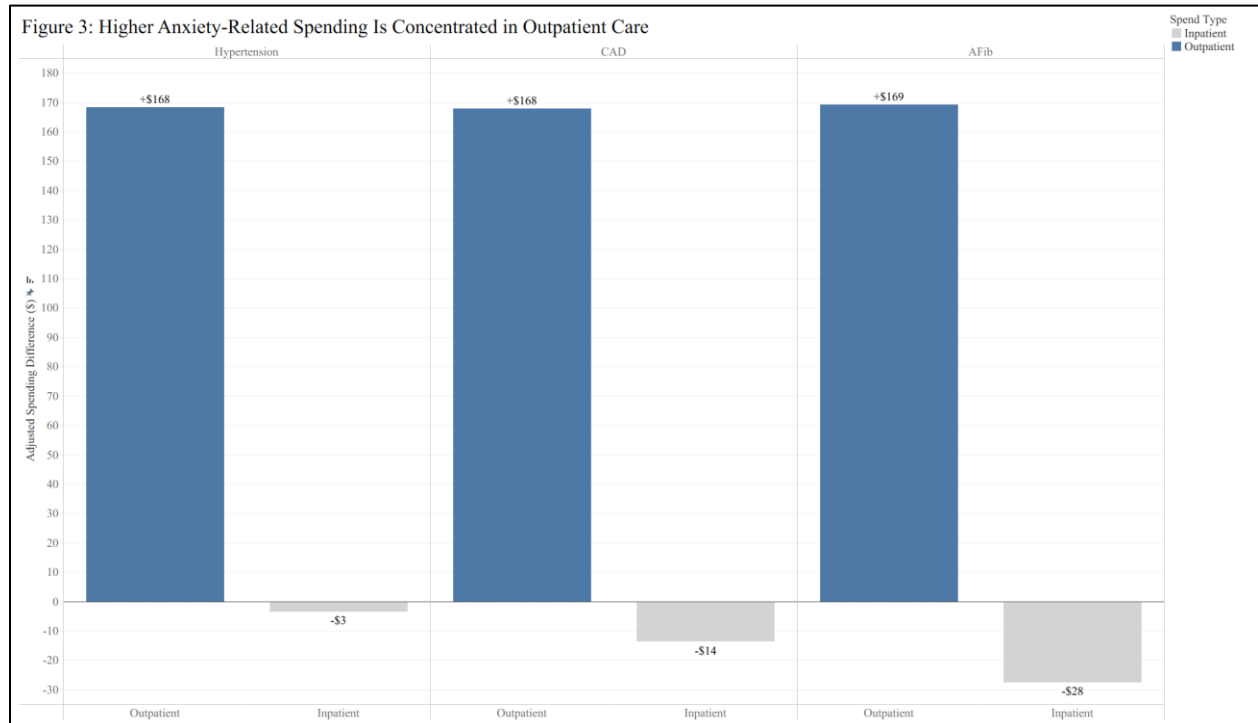


Figure 3: Adjusted spending differences by care setting and cardiovascular condition. Values represent the adjusted dollar difference in spending between patients with anxiety and those without anxiety (with anxiety minus without anxiety). Positive values indicate higher spending among patients with anxiety, while negative values indicate slightly lower spending. Differences are concentrated in outpatient care, while inpatient differences are minimal and slightly negative.

Separate two-part models for outpatient spending confirm this pattern. In Part 1, logistic regression models estimated the probability of any outpatient spending and showed that the odds were 43% to 58% higher among patients with anxiety (odds ratios: 1.43–1.58, all $p < 0.001$). In Part 2, Gamma generalized linear models estimated outpatient spending among patients with positive outpatient claims and showed that spending was approximately 10%–11% higher (cost ratios: 1.10–1.11, all $p < 0.001$). In contrast, anxiety had no significant effect on the probability of inpatient use (odds ratios near 1.0, all $p > 0.45$) or on inpatient spending among patients who were admitted. This pattern suggests that anxiety influences how patients interact with the healthcare system rather than increasing the likelihood of serious medical events. Patients with anxiety may seek care more frequently, require additional follow-up visits, or undergo more

diagnostic testing. These behaviors increase overall healthcare use without necessarily increasing inpatient admissions.

Geographic Variation

Figure 4 presents the distribution of state-level anxiety spending gaps. Each point represents one state-level estimate of the mean annual spending difference between patients with anxiety and those without anxiety. All 50 states and the District of Columbia (51 states total) show positive spending gaps, and the mean state-level gap is approximately \$3,011. Although the size of the gap varies across states, the lack of negative values indicates that the association between anxiety and higher spending is consistent across the United States. This suggests that the pattern is not driven by a small number of outlier states or by localized practice patterns but instead reflects a more systemic relationship between anxiety and healthcare use.

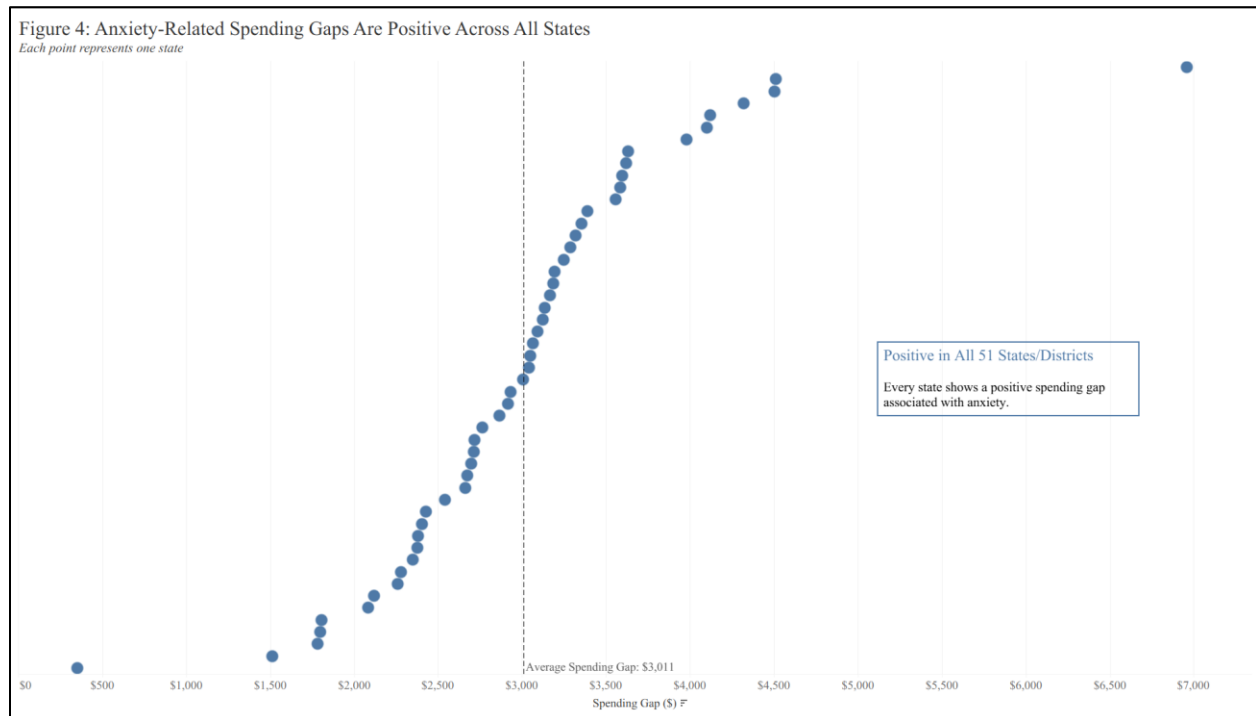


Figure 4: Distribution of state-level anxiety-associated spending gaps. Each point represents the mean annual Medicare spending gap between patients with anxiety and those without anxiety in a single state. All 51 states/districts show positive spending gaps. The mean state-level gap is approximately \$3,011 (SD: \$967, range: \$353–\$6,959). A one-sample t-test confirms the mean gap is significantly different from zero ($t = 22.24$, $p < 0.001$).

Although there is variation in the size of the gap across states (SD: \$967, range: \$353 to \$6,959), the consistent direction indicates that the relationship between anxiety and higher spending is not driven by regional factors. A one-sample t-test on the state-level spending gaps confirmed that the mean gap was significantly different from zero ($t = 22.24$, $p < 0.001$, 95% CI: \$2,739–\$3,283). The coefficient of variation was 0.32, indicating a relatively consistent gap across states. Additionally, interaction models found no significant difference in the anxiety spending gap across age groups (likelihood ratio test $p = 0.70$), although the effect did vary significantly across cardiovascular conditions ($p < 0.001$), with the largest adjusted gap observed for hypertension.

Discussion

This analysis shows that anxiety is consistently associated with higher Medicare spending across major cardiovascular conditions. Figure 1 demonstrates that the raw spending gaps are large, while Figure 2 shows that smaller but still meaningful differences remain after adjusting for demographic and clinical characteristics. Together, these figures suggest that the relationship between anxiety and higher spending is not fully explained by differences in age, sex, race, or comorbidity burden.

Figure 3 helps clarify the mechanism underlying this relationship. The adjusted spending gap is concentrated in outpatient care, while inpatient differences are minimal. This pattern suggests that anxiety influences how patients engage with outpatient services rather than increasing the likelihood of major medical events requiring hospitalization. Patients with anxiety may seek care more frequently, require additional follow-up visits, or undergo more testing and monitoring. While some of this care may be clinically appropriate, the consistency of the

outpatient pattern suggests that anxiety may contribute to a sustained increase in lower intensity healthcare use.

Figure 4 further strengthens the overall interpretation by showing that the anxiety associated spending gap is positive across all states examined. This consistency suggests that the relationship is not limited to a particular region or health system context. Instead, the association appears to be generalizable across the country.

From a policy perspective, these findings highlight the importance of integrating behavioral health into chronic disease management. If anxiety contributes independently to higher outpatient healthcare use, then screening, care coordination, and better access to mental health support may help reduce excess use while improving patient outcomes. As Medicare continues to move toward value based care, behavioral drivers of spending may become increasingly important targets for intervention.

Limitations

This study has several limitations. First, the use of observational claims data can limit causal interpretation, as unmeasured confounders may remain. Second, anxiety is identified based on diagnosis codes and may be underreported or misclassified. Third, spending measures do not capture clinical outcomes, limiting the ability to assess whether higher spending reflects better or worse care. Finally, the analysis does not account for variation in anxiety severity, which may influence healthcare use patterns.

Conclusion

Anxiety is associated with higher Medicare spending among patients with cardiovascular conditions, even after adjusting for demographic and clinical factors. The effect is driven primarily by outpatient care and is consistent across the United States. These findings suggest

that addressing anxiety as part of cardiovascular care may improve both efficiency and patient outcomes and represent an important opportunity for health system improvement.